

1. An Introduction to Data Warehousing

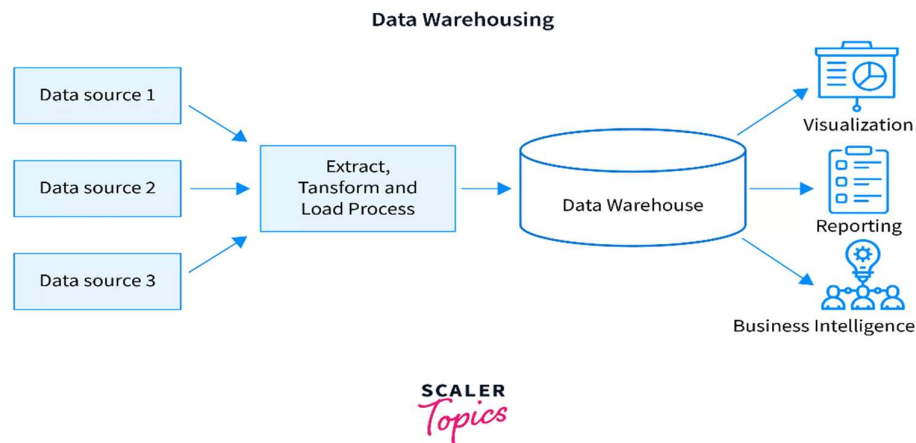
-- A **data warehouse** is a Subject oriented, integrated, time variant, non-volatile collection of data in support of management's system.

-- In simple words, it is a place where we store data for business purposes.

2. Purpose of Data Warehouse

- To integrate data from multiple sources
- To support **decision-making** by offering fast access to data
- To provide **historical data analysis**
- To improve **data quality and consistency**

3. Data Warehouse Architecture



The architecture has **3 main layers**:

1. **Data Source Layer**: Contains raw data from multiple sources (databases, applications, etc.)
2. **Data Staging Area (ETL)**: Extracts, Transforms, and Loads data

3. **Data Storage Layer:** Cleaned and transformed data is stored here (in fact and dimension tables)

ETL = Extract, Transform, Load

4. Operational Data Store (ODS)

- An **Operational Data Store** is used for real-time data integration.
- Unlike a data warehouse, it stores current data (not historical).
- It is used when businesses need up-to-date data quickly.

5. OLTP vs Data Warehouse Applications

Feature	OLTP (Online Transaction Processing)	Data Warehouse
Purpose	Day-to-day transactions	Data analysis
Data	Current, detailed	Historical, summarized
Operations	Read & write	Mostly read
Speed	Very fast	Slower
Users	Clerks, admins	Analysts, managers

6. Data Marts

- A **Data Mart** is a smaller version of a data warehouse.
- It focuses on a single department or business function (like Sales or HR).
- Data marts are faster and easier to use for specific teams.

7. Data Marts vs Data Warehouses

Data Warehouse	Data Mart
Data warehouse is a Centralised system.	While it is a decentralised system.
In data warehouse, lightly denormalization takes place.	While in Data mart, highly denormalization takes place.
Data warehouse is top-down model.	While it is a bottom-up model.
To built a warehouse is difficult.	While to build a mart is easy.
In data warehouse, Fact constellation schema is used.	While in this, Star schema and snowflake schema are used.
Data Warehouse is flexible.	While it is not flexible.

8. Data Warehouse Life Cycle

The life cycle includes:

1. **Requirement Gathering** – Understand business needs
2. **Design** – Create architecture, data models
3. **Development** – ETL process, schema design
4. **Testing** – Ensure data accuracy and quality
5. **Deployment** – Move to production
6. **Maintenance** – Monitor, optimize, and update